Gyires's permanent inequality for circulant matrices of order five

Overview of the result, its proof and its verification

Dev D. Goyal · October 2026 · github.com/dvgyl/gyires-circulant-5

The repository contains a proof that Gyires's permanent inequality holds for every doubly stochastic circulant matrix of order five, with its equality case, together with a stronger inequality for every nonnegative circulant of order five and a sharp quantitative margin. The proof is presented in a self-contained paper and formalized in Lean 4 with Mathlib; the formal proof uses only Lean's standard axioms, and all statements were also tested with exact computations written independently of the formal proof.

The result was obtained in an AI research effort directed by Dev D. Goyal. ChatGPT (OpenAI, GPT-6 with Pro reasoning), working autonomously from Dev D. Goyal's prompts, proposed the problem, found the proof and wrote the Lean formalization. Claude Opus 5.5, used with Claude Code, checked the work independently and wrote the paper in its present form. Dev D. Goyal has reviewed the proofs; this review is not an independent expert validation. The paper has not been peer reviewed.

1. The conjecture

A real square matrix is doubly stochastic if its entries are nonnegative and its row and column sums are all 1; the set of such $n \times n$ matrices is Ω_n, and J_n is the matrix all of whose entries are $1/n$. B. Gyires conjectured in 1978 that every $A \in \Omega_n$ satisfies

$$\frac{4\,\mathrm{per}(A)^2}{\mathrm{per}(AA^{\mathsf{T}}) + \mathrm{per}(A^{\mathsf{T}}A) + 2\,\mathrm{per}(A^2)} \;\geq\; \frac{n!}{n^n},$$

with equality only at $A = J_n$ (Conjecture 20 in the survey of Cheon and Wanless, Linear Algebra Appl. 403, 2005). The case $n = 2$ is elementary, the inequality for $n = 3$ was proved by D. K. Chang in 1990, and the case $n = 4$ was proved by C. Ozols in 2026 with a computer-assisted proof formalized in Lean (not peer reviewed). The conjecture is open for $n \geq 5$.

2. The result

A circulant matrix of order five has entries $c_{ij} = w_{j-i}$, indices modulo five. The doubly stochastic circulants are the convex combinations of the five powers of the cyclic permutation matrix: a four-dimensional simplex inside the sixteen-dimensional polytope Ω_5; this simplex contains matrices with zero entries. For every circulant C of order five with nonnegative weights (the entries of its first row):

- the strong inequality $24\,\mathrm{per}(CC^{\mathsf{T}}) \leq 625\,\mathrm{per}(C)^2$ holds, with equality exactly when all five weights are equal;
- if $z_1, \ldots, z_4$ are the gaps between the sorted weights, then $625\,\mathrm{per}(C)^2 - 24\,\mathrm{per}(CC^{\mathsf{T}}) \geq 484\,(z_1^{10} + z_2^{10} + z_3^{10} + z_4^{10})$, and 484 is best possible;
- Gyires's inequality holds for C, with equality exactly when all weights are equal. In particular, for every doubly stochastic circulant of order five the ratio of Section 1 is at least $5!/5^5 = 24/625$, with equality exactly at J_5.

3. The proof

Every circulant is normal, and $\operatorname{per}(C^2) \le \operatorname{per}(CC^{\mathsf{T}})$ by a comparison of the form $2\operatorname{per}(MN^{\mathsf{T}}) \le \operatorname{per}(MM^{\mathsf{T}}) + \operatorname{per}(NN^{\mathsf{T}})$, so Gyires's inequality follows from the strong inequality. The defect $625\operatorname{per}(C)^2 - 24\operatorname{per}(CC^{\mathsf{T}})$ is a homogeneous polynomial of degree ten in the weights, invariant under the twenty affine changes of indices $i \mapsto ai + b$ modulo five. These reduce the 120 orderings of the weights to six. For each of the six, the defect, written in the smallest weight and the gaps between consecutive sorted weights, has 1001 coefficients, all nonnegative integers: 996 are positive, the least being 484, and five are zero. This gives the inequality, the margin and the equality case at once.

4. Verification

The Lean project builds from source with Lean 4.35.0 and Mathlib v4.35.0. Ten principal theorems are restated with Mathlib's definitions only. Comparator, a Lean tool that checks that a proof proves exactly a statement given in a separate file, confirms that the proved theorems have exactly these statements, and the proofs are accepted by Lean's kernel and by the independent NanoDa and con-ron kernels. The chamber certificate is checked in Lean coefficient by coefficient, by the `ring` and `positivity` tactics, without evaluation by compiled code. A separate PARI/GP script recomputes every permanent from the definition, recomputes the chamber polynomials for all 120 orderings, and tests every statement of the paper.

5. The paper

Gyires's permanent inequality for nonnegative circulant matrices of order five.

Conjecture 20 of Cheon and Wanless for doubly stochastic circulants of order five, with its equality case; the strong inequality and its sharp margin for nonnegative circulants. *(Special case proved.)*

Paper · Lean statements · Checks

The result concerns circulant matrices of order five only; Gyires's conjecture for all of Ω_5 remains open. The searches for earlier work were limited, and no claim of novelty or priority is made.